“LearnMate AI: Personalized Learning Roadmap Generator”

IBM University Engagement Project Submission

Project Title – LearnMate AI: Personalized Learning Roadmap Generator

Domain of Project – Education / EdTech

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Project overview

LearnMate AI is an Education / EdTech application that generates personalized learning roadmaps from a learner's education, current skills, career interest, daily study hours, and preferred learning duration.

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Problem Statement

Problem Statement: Personalized Learning Roadmaps

The Challenge

Learners often know the career they want but are unsure what to learn, in what order, and how much time to allocate. Generic roadmaps may not reflect a learner’s education level, existing skills, available study hours, or target timeline. This makes it difficult to turn a career goal into a realistic, structured learning plan.

The Objective

Build LearnMate AI, a personalized learning pathway generator that converts a learner profile into a clear roadmap.

- Capture education, current skills, career interest, daily study hours, and target duration.
- Generate a month-by-month plan with weekly goals and practical projects.
- Keep the roadmap aligned with the learner’s selected duration and study availability.
- Present the generated plan in a simple, accessible interface.

Proposed Solution

Proposed Solution: LearnMate AI

LearnMate AI is a Streamlit application that creates a personalized learning roadmap using an existing IBM watsonx.ai deployment.

The learner submits a profile containing their name, education, current skills, career interest, daily study hours, and learning duration. The application sends these values in a structured prompt to the deployed model.

IBM watsonx.ai returns a roadmap organized month by month, with weekly learning goals, practical projects, and relevant tools or technologies. The app displays the learner profile alongside the generated plan.

The application also validates the response against the selected duration. It trims schedule sections that exceed the requested timeline and rejects a roadmap that still falls outside the duration.

The result is a practical starting point for learners to plan focused, goal-oriented self-study.

Technology Used

- Streamlit

Builds the interactive web interface, student profile form, and roadmap display.

IBM watsonx.ai Deployment Inference API

Receives the personalized prompt and generates the learning roadmap.

IBM Granite / Deployed Foundation Model

Provides natural-language generation through the configured watsonx.ai deployment. The exact model identifier is deployment-configured and is not specified in the project files.

IBM Cloud IAM

Exchanges the IBM Cloud API key for a short-lived bearer access token at runtime.

Python

Implements prompt construction, API requests, response handling, and roadmap validation.

Requests + python-dotenv

Requests handles HTTPS API calls; python-dotenv loads local environment configuration.

Environment variables

WATSONX_API_KEY, WATSONX_URL, and WATSONX_DEPLOYMENT_ID configure secure access to the deployment.

Application Components Used

1) Student Profile Form – Collects the learner name, education level, current skills, career interest, daily study hours, and learning duration.

2) Streamlit Form Controls – Select boxes and text inputs make profile entry simple and structured.

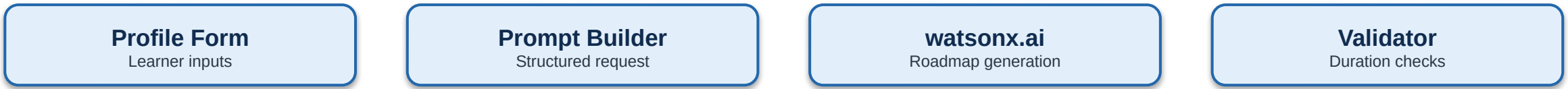
3) Prompt Builder – Combines the six profile values into a clear instruction for roadmap generation.

4) IBM watsonx.ai API Call – Sends the prompt to the configured deployment using an IBM IAM bearer token.

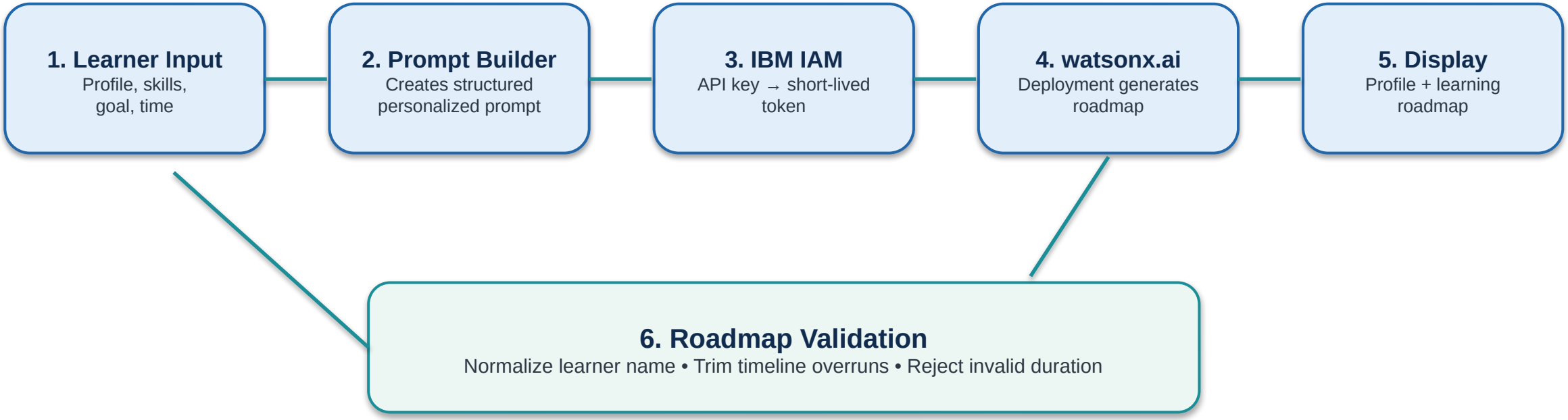
5) Roadmap Response Handler – Extracts the generated assistant content from the API response.

6) Duration Validation – Checks month and week references, trims overrun sections, and rejects a roadmap that still exceeds the selected duration.

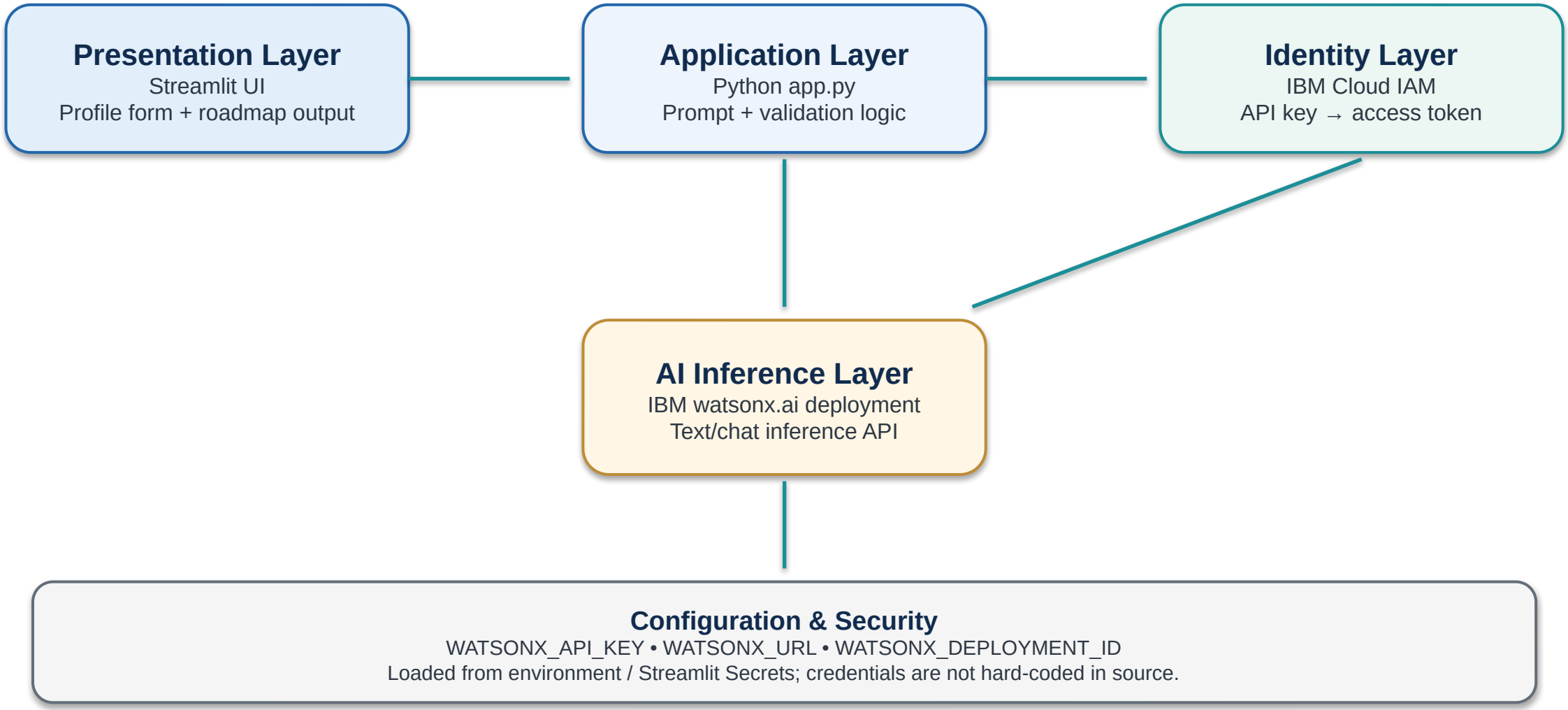
7) Student Profile & Roadmap Display – Shows submitted details and the generated learning plan in the Streamlit interface.



LearnMate AI Workflow



Architecture Blueprint

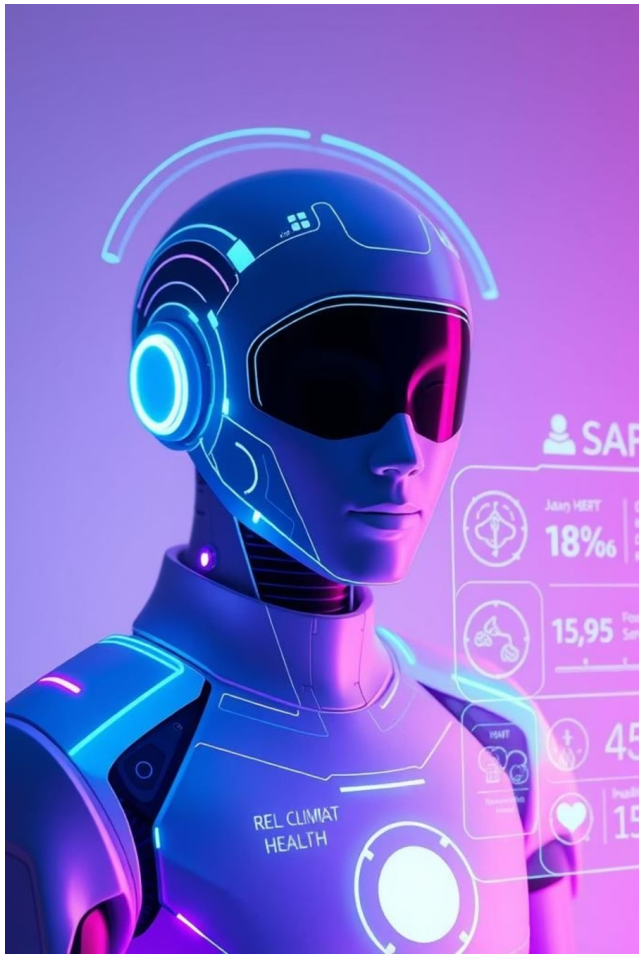


Role of Generative AI in LearnMate AI

The deployed IBM watsonx.ai model is the generation engine behind LearnMate AI.

- Contextual generation: uses the learner’s education, skills, career goal, study hours, and duration as prompt context.
- Structured planning: is instructed to organize the roadmap month by month and include weekly goals and practical projects.
- Personalization: adapts the learning plan to the learner profile rather than returning a single generic pathway.
- Timeline awareness: receives the selected duration as an explicit constraint.
- Human review: the generated roadmap is presented to the learner as guidance; learners should review and adapt it to their needs.

The current code uses one deployed inference endpoint. The repository does not implement a multi-agent orchestration framework, so this presentation describes the solution as a personalized generative-AI application rather than a multi-agent system.



Usage & Validation

Token Usage – 8,975 tokens reported in the submitted project PDF

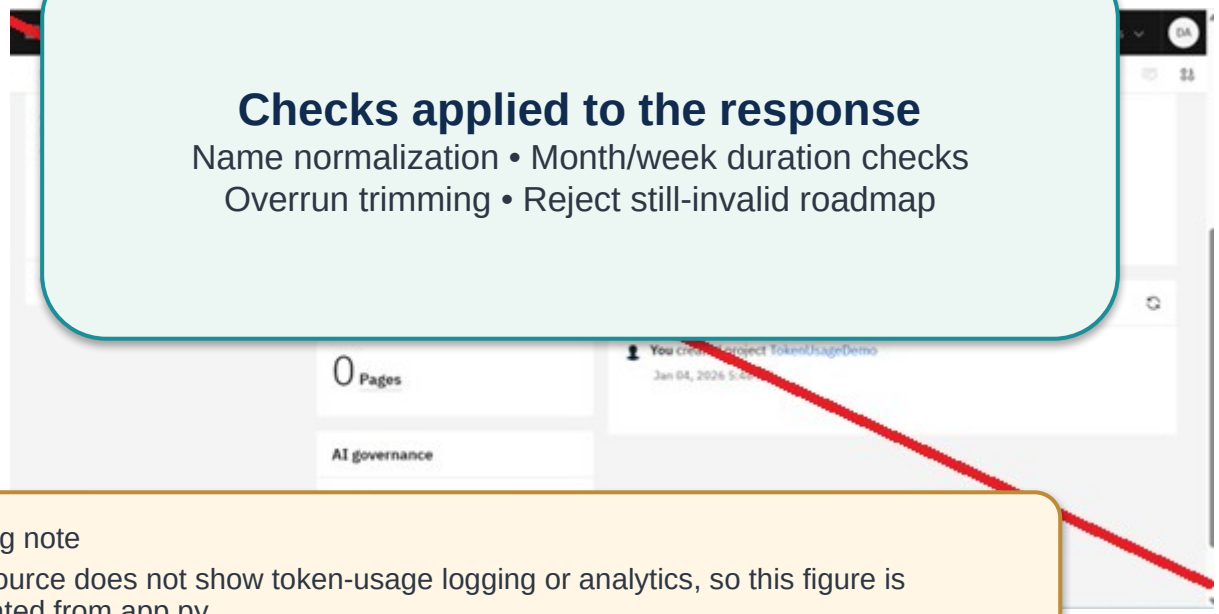
Before generation

Input sent to the model

Six learner fields are inserted into a structured prompt:
name • education • skills • career • study hours • duration

Checks applied to the response

Name normalization • Month/week duration checks
Overrun trimming • Reject still-invalid roadmap



Reporting note

The submitted project PDF reports 8,975 tokens used. The supplied application source does not show token-usage logging or analytics, so this figure is presented as the reported project value rather than a metric independently calculated from app.py.

Novelty and Uniqueness

LearnMate AI focuses on turning a learner profile into a practical, time-bounded learning roadmap.

Profile-Based Personalization – Uses education, current skills, career interest, daily study hours, and target duration to shape the generated plan.

Structured Roadmap Output – Prompts the model to provide month-by-month learning, weekly goals, and practical projects.

Duration Guardrails – Checks the generated roadmap against the selected learning period, trims detected overruns, and rejects a still-invalid result.

Secure Credential Handling – Reads IBM Cloud credentials and deployment settings from environment variables or Streamlit Secrets rather than embedding them in application code.

Simple Learner Experience – Uses a Streamlit form to collect inputs and display both the student profile and generated roadmap.

Deployment-Based AI – Connects to an existing IBM watsonx.ai deployment through the inference API, keeping model hosting separate from the app.

GitHub Repository

GitHub profile provided:

https://github.com/priyakumari0202

Note: This is the GitHub profile URL. If your LearnMate AI project is in a separate repository, replace this with the repository URL before submission.

Project files to include in the repository:

- README.md – setup and usage instructions
- app.py – Streamlit application and watsonx.ai integration
- requirements.txt – Python dependencies
- .gitignore – excludes local secrets and development files

Before submission:

1. Confirm the project repository is public.
2. Check that all required project files are uploaded.
3. Do not commit .env files, API keys, or other credentials.
4. Add the final presentation and required problem-statement PDF.

Future Scope

1. Learning Progress Tracking

Allow learners to mark weekly topics and projects as complete, then compare their progress with the generated roadmap.

2. Adaptive Roadmap Updates

Regenerate or adjust the learning plan when a learner’s available study hours, current skills, career goal, or target duration changes.

3. Learning Resource Integration

Add curated links to courses, documentation, tutorials, and practice platforms for each roadmap stage.

4. Skill Assessment

Introduce quizzes or self-assessments to identify knowledge gaps and prioritize the next topics.

5. Multilingual and Voice Interaction

Make roadmap creation more accessible through regional-language prompts and voice-based input.

These are proposed enhancements; they are not represented as implemented features in the current project source.

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Priya kumari

for the completion of

Getting Started with Artificial Intelligence

(PLAN-E624C2604060)

According to the Your Learning Builder - Plans system of record

Completion date: 17 Oct 2026 (GMT)

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Priya kumari

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 22 Oct 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.